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import java.io.*;
import java.util.ArrayList;

public class GradeAnalyzer {

    public static void main(String[] args) {

        String outputFile = null;
        if (args.length > 0) {
            outputFile = args[0];
        }
        if (outputFile == null || outputFile.length() == 0) {
            outputFile = "report.txt";
        }

        // Step 1: read scores from file
        String fileName = "scores.txt";
        ArrayList<Integer> scores = readScores(fileName);

        // Step 2: calculate statistics
        ArrayList<Integer> cleanedScores = new ArrayList<>();
        int errCount = 0;
        Integer min = Integer.MAX_VALUE;
        Integer max = Integer.MIN_VALUE;
        for (int i = 0; i < scores.size(); i++) {
            Integer score = scores.get(i);

            if (score < 0) {
                errCount++;
                continue;
            }
            cleanedScores.add(score);

            if (score < min) {
                min = score;
            }

            if (score > max) {
                max = score;
            }
        }
        double avg = calculateAverage(cleanedScores);

        // Step 3: write and print report
        writeReport(cleanedScores, avg, max, min, errCount, outputFile);
    }

    // Returns a list of valid scores read from the file
    public static ArrayList<Integer> readScores(String filename) {
        ArrayList<Integer> scores = new ArrayList<>();

        try (BufferedReader reader = new BufferedReader(new FileReader(filename))) {
            String line;
            while ((line = reader.readLine()) != null) {
                try {
                    Integer score = Integer.valueOf(line);
                    scores.add(score);
                } catch (NullPointerException | NumberFormatException e) {
                    System.out.println("Current line text input cannot be turned
into a number: \"" + line + "\"");
                }
            }
        }
    }
}

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        scores.add(-1);
    }
}
} catch (IOException e) {
    System.out.println("Could not read file: " + e.getMessage());
}

return scores;
}

// Returns the average of a list of scores, or 0.0 if the list is empty
public static double calculateAverage(ArrayList<Integer> scores) {
    if (scores.isEmpty()) return 0.0;

    int totalScores = 0;
    for (int i = 0; i < scores.size(); i++) {
        totalScores += scores.get(i);
    }

    return (double) (totalScores / scores.size());
}

// Writes and prints the report
public static void writeReport(ArrayList<Integer> scores,
                                double avg, int high, int low, int errCount,
                                String outputFile) {
    try (BufferedWriter writer = new BufferedWriter(new FileWriter(outputFile))) {
        writer.write("=== Grade Analysis Report ===");
        System.out.println("=== Grade Analysis Report ===");
        writer.newLine();
        writer.write("Total scores processed: " + scores.size());
        System.out.println("Total scores processed: " + scores.size());
        writer.newLine();
        writer.write("Invalid lines skipped: " + errCount);
        System.out.println("Invalid lines skipped: " + errCount);
        System.out.println();
        writer.newLine();
        writer.newLine();
        writer.write("Average score: " + avg);
        System.out.println("Average score: " + avg);
        writer.newLine();
        writer.write("Highest score: " + high);
        System.out.println("Highest score: " + high);
        writer.newLine();
        writer.write("Lowest score: " + low);
        System.out.println("Lowest score: " + low);
        System.out.println();
        writer.newLine();
        writer.newLine();

        int nineties = 0;
        int eighties = 0;
        int seventies = 0;
        int sixties = 0;
        int belowSixty = 0;
        for (int i = 0; i < scores.size(); i++) {
            int score = scores.get(i);
            if (score >= 90) {
                nineties++;
            } else if (score >= 80) {
                eighties++;
            }
        }
    }
}

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    } else if (score >= 70) {
        seventies++;
    } else if (score >= 60) {
        sixties++;
    } else {
        belowSixty++;
    }
}

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writer.write("Grade distributions:");
System.out.println("Grade distributions:");
writer.newLine();
writer.write("    A (90-100): " + nineties);
System.out.println("    A (90-100): " + nineties);
writer.newLine();
writer.write("    B (80-89): " + eighties);
System.out.println("    B (80-89): " + eighties);
writer.newLine();
writer.write("    C (70-79): " + seventies);
System.out.println("    C (70-79): " + seventies);
writer.newLine();
System.out.println("    D (60-69): " + sixties);
writer.newLine();
System.out.println("    F (below 60): " + belowSixty);
writer.newLine();
} catch (IOException e) {
    System.out.println("Could not write file: " + e.getMessage());
}
}
}

```